MIL-B-14316(ORD)
6 June 1956
SUPERSEDING
MIL-B-20361
28 November 1951
MIL-B-20392
4 December 1951
PA-PD-524
20 January 1955

MILITARY SPECIFICATION

BOOSTER, M21A4 AND M25
LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of booster, designated as Booster, M21A4 and M25, for use in high explosive loaded, time-fuzed and impact-fuzed projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-A-351 - Aluminum Alloy 17S Bars, Rods, and Wire (Rolled and Drawn)

MILITARY

JAN-P-223 - Powder, Black
MIL-G-2550 - General Specification For Ammunition Except Small Arms Ammunition
MIL-D-2493 - Detonators, Loading, Assembly and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-350 - Jolt Test for Use in Production of Fuzes

FSC 1320

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DRAWINGS

ORDNANCE CORPS

73-2-154 - Booster, M21A4 Assembly
73-2-213 - Booster, M25 Assembly
81-3-35 - Machine Jumble Testing
81-3-37 - Fixture Jumble Testing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Boosters.-The booster shall comply with all requirements specified on Drawings (dwg.) 73-2-154 or 73-2-213, as applicable, and with all requirements specified in applicable specifications.

3.3 Check test for possible deterioration of M17 detonator.-The detonators shall be subjected to and must satisfactorily pass the check test for possible deterioration specified in Specification MIL-D-2493 prior to assembly into the booster if more than the time specified in Specification MIL-D-2493 has elapsed.

3.4 Jolt and jumble.-The booster shall withstand the jolt test specified in Standard MIL-STD-350 except that the requirement specified in 5.6 of MIL-STD-350 shall be deferred until both the jolt and jumble tests have been completed. (See 6.2). After completion of the jolt test, the boosters used in the jolt test shall withstand 3,600 complete revolutions in the jumble machine without having any explosive elements explode, booster come apart, booster partially or fully arm, inner parts broken, deformed, or displaced in such a manner as to make the booster unsafe to handle or dangerous to fire when tested as specified in 4.7.

3.5 Static functioning.-The booster shall detonate satisfactorily when fired statically as specified in 4.8.

3.6 Safety and arming.

3.6.1 Safety.-The booster shall not fail to detonate the projectile filler satisfactorily when tested as specified in 4.9.1.1.

3.6.2 Arming.-The booster shall not fail to detonate the projectile filler satisfactorily when tested as specified in 4.9.1.2.

3.6.3 Premature bursts.-There shall be no premature burst or evidence thereof when tested as specified in 4.9.1.1 and 4.9.1.2. A premature burst is defined in the case of point detonating fuzes as a functioning of the shell in the bore of the gun or in flight before impact, or in the case of time fuzes as a functioning in flight prior to one second from the time of discharge.

3.7 Workmanship.-All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent used.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of all boosters loaded and assembled in one unchanged process by one manufacturer under one contract in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain:

a. metal parts of one manufacturer bearing the same interfix lot number.

b. detonators of one manufacturer bearing the same interfix lot number.

c. tetryl from not more than two lots.

d. boosters of one designation only.

4.2 Sampling.

4.2.1 For non-destuctive inspection.-Unless otherwise specified herein, sampling for non-destructive inspection shall be conducted in accordance with Standard MIL-STD-105.

4.2.2 For specific gravity test.-Ten boosters shall be selected from each lot for test.

4.2.3 For jolt and jumble test.-Twelve boosters shall be selected from each lot for test.

4.2.4 For static functioning test.-Twenty-five boosters shall be selected from each lot for test.

4.2.5 Safety and arming.

4.2.5.1 For safety test.-Ten boosters shall be selected from each lot for test.

4.2.5.2 For arming test.-Ten boosters shall be selected from each lot for test.

4.3 Inspection.

4.3.1 General.-Inspection shall be as specified in Specification MIL-G-2550 and as specified herein.

4.3.2 Components.-It shall be ascertained that prior to assembly all components of sub-assemblies of the boosters procured under separate specifications or drawings have been inspected, tested, and accepted in accordance with their respective specifications or drawings.

4.3.3 Assembling and loading.-The assembling of the boosters and the loading of the tetryl shall be observed and inspected for conformance to this specification by a Government inspector at the place of manufacture.

4.3.4 Packing and marking.-It shall be ascertained that the packing and marking of the boosters shall conform to this specification.

4.3.5 Classification of defects.-Inspection of the boosters shall be as follows:

4.3.5.1 Body assembly, prior to crimping rotor cover (see dwg. 73-2-156 covering a detail of dwg. 73-2-154 or 73-2-213 as applicable).

Categories and defects	Gage or other method of inspection
Critical	
a. Rotor not completely locked in unarmed position	Visual
b. Detonator disc ruptured exposing explosive	Visual
c. Detonator above flush	Visual

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Major	AQL 1.00 percent	
1.	Arming test failure	Approved testing device (see 4.6.1)
2.	Detonator missing	Visual
3.	Detonator inverted	Visual
4.	Rotor inverted	Visual
5.	Rotor lock pin missing	Visual
6.	Rotor lock pin lock missing	Visual
7.	Rotor lock pin and rotor lock pin lock in reversed location	Visual
8.	Detonator loose or not having a full 340° crimp and sufficiently formed to anchor the detonator firmly	Visual-Gage
9.	Detonator cushion missing	Visual
10.	Booster closing cup protruding in rotor cavity	Visual-Straight Edge
11.	Booster closing cup charge loose, cracked or chipped	Visual
Minor	AQL 1.50 percent	
41.	Non-arming test failure	Approved testing device (see 4.6.2)
42.	Foreign matter in rotor cavity	Visual
43.	Shellac on underside of rotor cover assembly	Visual
44.	Shellac covering more than 1/4 of the flash hole in the rotor cover	Visual

4.3.5.2 Assembly, prior to assembly of booster cup (see dwg. 73-2-154 or 73-2-213 as applicable).

	Categories and defects	Gage or other method of inspection
Critical		
a.	Solid tetryl bridges from one booster cup thread to another	Visual
Major	AQL 0.25 percent	
1.	Booster pellet missing	Visual
2.	Booster closing cup assembly-missing	Visual
Minor	AQL 1.50 percent	
41.	Booster closing cup explosive above flush	Gage
42.	Booster closing cup explosive below bottom surface of body; maximum (max.)	Gage
43.	Booster closing cup not having a full 360° crimp to sufficiently anchor the cup firmly	Visual
44.	Foreign matter in booster cup	Visual

Major		
1.	Specific gravity of pellet	Witnessed by the inspector (see 4.5)
2.	Binder-lubricant content, max.	Certification (see 4.4)

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4.3.5.3 Assembly (see dwg. 73-2-154 or 73-2-213 as applicable).

Categories and defects	Gage or other method of inspection
Critical	
a. Rotor cover missing	Visual
Major	AQL 0.40 percent
1. Closing screw or body plug missing	Visual
2. Rotor cover stake or crimp missing	Visual
3. Threads damaged	Visual-Gage
4. Rotor not free	Manual
Minor	AQL 4.00 percent
41. Length, maximum	Gage
42. Concentricity of booster cup with body diameter above cup	Gage
43. Booster cup contacting shoulder of body	Visual
44. Booster cup loosens when an unsrewing torque of 35 inch-pounds is applied	Approved testing device
45. Booster cup cracked or split (ignoring slight crack extending from stake mark to open end)	Visual
46. Rotor cover disc missing or punctured	Visual
47. Foreign matter	Visual
48. Marking misleading or unidentifiable	Visual
49. More than one rotor cover	Visual
50. Booster pellet moves when assembly is shaken (disassemble if loose pellet is suspected)	Manual
51. Rotor cover loose	Manual

Categories and defects	Gage or other method of inspection
Major	
1. Jolt test failure	Witnessed (see 4.7) by the inspector
2. Jumble test failure	Witnessed (see 4.6) by the inspector
3. Static functioning test failure	Witnessed (see 4.8) by the inspector

4.3.5.4 Packing set-up box (see dwg. 76-7-644 covering a detail of dwg. 73-2-154 or 76-7-743 covering a detail of 73-2-213 as applicable).

Categories and defects	Gage or other method of inspection
Major	AQL 0.25 percent
1. Assembly damaged so that contents are exposed or liable to become exposed	Visual
2. Sealing strip incomplete, badly wrinkled, or not in contact with surface of containers for at least 1/4 inch around entire circumference	Visual
Minor	AQL 1.50 percent
41. Contents move upon shaking	Manual
42. Marking misleading or unidentifiable	Visual
43. Label missing	Visual

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4.3.5.5 Box, packing (see dwg. 76-1-1239 covering a detail of 73-2-154 or 76-7-742 covering a detail of 73-2-213 as applicable).

	Categories and defects	Gage or other method of inspection
Major	AQL 0.65 percent	
1.	Board broken	Visual
2.	Hardware missing, broken or loose	Visual-Manual
3.	Contents exposed	Visual
4.	AIC symbol misleading or unidentifiable	Visual
5.	Strapping missing, broken or loose	Visual-Manual
Minor	AQL 1.50 percent	
41.	Hardware improperly engaged	Visual-Manual
42.	Contents move when shaken	Manual
43.	Strapping improperly assembled	Visual-Manual
44.	Car seal missing, unsealed or improperly positioned	Visual-Manual
45.	Handle missing or insecure	Visual-Manual
46.	Marking misleading or unidentifiable	Visual

4.4 Determination of binder-lubricant content of tetryl.-(See 6.1) A minimum of one sample of tetryl with a maximum of five samples required from each lot shall be selected for determination of the binder-lubricant content in each lot of boosters. Approximately eight ounces shall be selected and determined as follows: An accurately weighed portion of five grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four-10-milliliter (ml.) portions of acetone. Each portion shall be maintained in contact with the residue for approximately five minutes before applying suction. Any lumps which may persist shall be broken up. The sides of the crucible shall be washed down with each 10-ml. portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (C) to 105° Centigrade (C) for ½ hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10-ml. portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

4.5 Determination of specific gravity of booster pellet.

4.5.1 Procedure.-The representative booster pellets before assembling the booster shall be tested as follows:

The representative booster pellets shall be cleaned so as to leave no sharp corners and shall then be accurately weighed on a laboratory balance. To determine the displacement, an apparatus (see figure 1) consisting of a glass beaker with ground top, a piece of flat glass of sufficient size to completely cover the top of the beaker, and a semihard rubber perforated plate, pierced at each corner with sharp pointed pins so that about one-half inch of pin protrudes from each side shall be used. The beaker shall be

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filled with clean mercury, the rubber plate inserted, and the mercury leveled off by pressing down with the flat glass until it is even with the top of the beaker. The representative booster pellets shall be placed beneath the prongs of the rubber plate and the mercury leveled as before. The excess mercury shall be caught in a clean bowl. The mercury displaced by the pellets shall be weighed and the specific gravity of the representative pellets shall be calculated as follows to determine compliance with the requirement specified on the applicable drawing:

$$\text{Specific gravity} = \frac{13.54W}{W_1}$$

where:

W = weight of booster pellet in grams
W_1 = weight of displaced mercury in grams

4.5.2 Rejection.-If the specific gravity of one or more booster pellets fails to comply with the requirements specified on the applicable drawing, the lot shall be rejected without further test.

4.6 Test of arming mechanism.

4.6.1 Arming.

4.6.1.1 Procedure.-Prior to staking of cover and disc assembly, the representative booster body assemblies selected in accordance with 4.3.5.1 shall be placed in a spinning machine equipped with a spindle as shown on figures 2 and 3. The centrifugal pin lock pin shall be depressed to seat under the web of the counterbore to permit free movement of the centrifugal pin. The representative body assembly shall then be placed in the spinning machine and rotated at the arming speed specified on the drawing in such a manner that centrifugal force alone accomplishes arming. Observation shall be made to determine compliance with the arming requirement specified on the applicable drawing. (Arming of the booster shall be considered to have taken place when the detonator portion of the rotor is aligned with the flash hole and the rotor is stopped against the stop pin.)

4.6.1.2 Retest and rejection.-If the number of boosters failing to comply with the arming requirements specified on the applicable drawing exceeds the acceptance number, but is less than the rejection number specified by the appropriate sampling plan, a second sample of double the number of boosters used in the first sample shall be tested. If the number of failures observed in the first sample equals or exceeds the rejection number specified by the appropriate sample plan the lot shall be rejected. If, at any time, during the test of the second sample the number of failures equals or exceeds the rejection number specified by the appropriate sampling plan the lot shall be rejected without further test.

4.6.2 Non-arming.

4.6.2.1 Procedure.-Following the arming test, the rotors shall be replaced in the non-armed position and the representative booster body assemblies selected in accordance with 4.3.5.1, shall be replaced in the spinning machine in the same manner as specified in 4.6.1 and retested at the non-arming speed specified on the drawing. Observation shall be made to determine compliance with the non-arming requirement specified on the applicable drawing. (The non-arming requirement shall be considered to have been complied with if the rotor has not released and has remained in the fully non-armed position.) Immediately subsequent to this test,

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the centrifugal pin lock pin shall be returned to its normal or non-armed position. (The centrifugal pin shall be in the pin recessed slot of rotor so that the detonator face of rotor is adjacent to the stop pin.) The axis of the spindle of the spinning machine shall be vertical, and the spindle shall be of the continuously operating type. The spindle shall be accurately aligned and balanced, and the spinning machine shall be rigidly mounted on a solid base, and shall be so adjusted that in the arming test, the speed of rotation of the spindle shall not exceed by more than 1 percent the arming speed specified on the drawing. In the non-arming test, the speed of rotation shall be not more than 1 percent less than the non-arming speed specified on the drawing. The direction of roation shall be counterclockwise.

4.6.2.2 Retest and rejection.-If the number of boosters failing to comply with the non-arming requirement specified on the applicable drawing exceeds the acceptance number, but is less than the rejection number specified by the appropriate sampling plan a second sample of double the number of boosters used in the first sample shall be tested. If the number of failures observed in the first sample equals, or exceeds the rejection number specified by the appropriate sampling plan the lot shall be rejected. If, at any time, during the test of the second sample the number of failures equals or exceeds the rejection number specified by the appropriate sampling plan the lot shall be rejected without further test.

4.7 Jolt and jumble

4.7.1 Procedure.-The representative boosters shall be assembled in an adapter as shown in figure 4, and subjected to the jolt test as specified in Standard MIL-STD-350. After completion of the jolt test, the representative boosters used in the jolt test shall be subjected to the jumble test. The representative boosters shall be assembled in jumble fixture in accordance with dwg. 81-3-37,and a suitable adapter and placed, oneat a time, in jumble machine dwg. 81-3-35. The machine shall be given 3,600 complete revolutions at the speed specified on dwg. 81-3-35. The boosters shall be examined upon completion of this test. The representative boosters shall be disassembled after both the jolt and jumble tests are completed.

4.7.1.2 Rejection.-If one or more boosters fail to comply with the requirements of 3.4 the lot shall be rejected without further test.

4.8 Static functioning.

4.8.1 Procedure.-The rotor assembly of the representative boosters shall be armed and assembled for electric firing as shown on figures 5 and 6. The assembly shall be placed on an aluminum cylinder two inches in diameter by three inches in height as shown on figure 7, with the booster cup mounted on the aluminum cylinder. The electric squib shall be fired and observation made for satisfactory functioning: Satisfactory functioning shall be indicated when the cylinder compares with figure 7.

4.8.2 Rejection.-If one or more boosters fail to comply with the requirements of 3.5 the lot shall be rejected without further test.

4.9 Safety and arming.-The following tests shall be performed at a Government Proving Ground:

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4.9.1 Procedure.

4.9.1.1 Safety.-The representative boosters shall be assembled to standard time fuzes for which the booster is standard, using fuzes from an accepted lot. The fuze-booster assemblies shall be assembled to a standard high-explosive shell with a nominal weight of 15 pounds and fired with fuzes set for 5 seconds time of flight, from a 76-mm gun with a chamber pressure of 43,000 ± 500 pounds per square inch.

4.9.1.2 Arming.-The representative boosters shall be assembled to standard point detonating fuzes for which the booster is standard, using fuzes from an accepted lot. The fuze booster assemblies shall be assembled to a standard high explosive shell and fired with fuzes set "super-quick" for ground impact functioning at a range of approximately 3,000 yards, from a 155-mm M1 Howitzer, at a muzzle velocity of 625 feet per second and a rotational velocity of 2880 revolutions per minute.

4.9.2 Retest and rejection.-If in the phase being test (i.e., safety or arming) no premature burst occurs and if one but not more than two boosters fail to detonate the projectile filler satisfactorily, the test shall be repeated on 20 boosters. If, at any time during the test a premature burst occurs or if more than two boosters fail to detonate the projectile filler satisfactorily, or if during the retest a premature brust occurs or if the total number of boosters which fail to detonate the projectile filler satisfactorily on the original plus retest exceeds a total of two, the lot shall be rejected without further test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, marking.-Packaging, packing, labeling, and marking shall be as specified on the applicable drawing.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Boosters required (see 1.1).

(c) Certification required to guarantee that the binder-lubricant content of the tetryl specified in 4.4 has been complied with.

6.2 Although the scope for use of this standard applies to fuzes only, current practice indicates that it may also apply to boosters.

6.3 If boosters are assigned for assembly in specified fuzes, separate jolt and jumble and separate ballistics tests can be omitted and combined with the corresponding acceptance test of the fuze.

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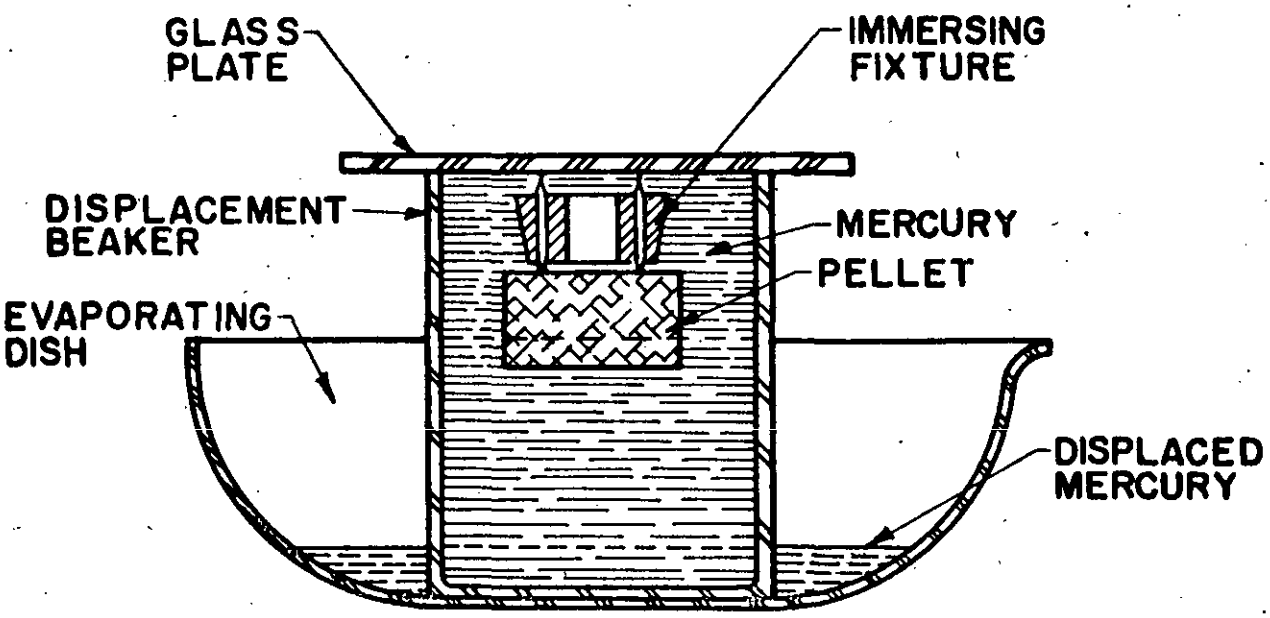


FIGURE 1- EQUIPMENT SET UP FOR DETERMINING SPECIFIC GRAVITY.

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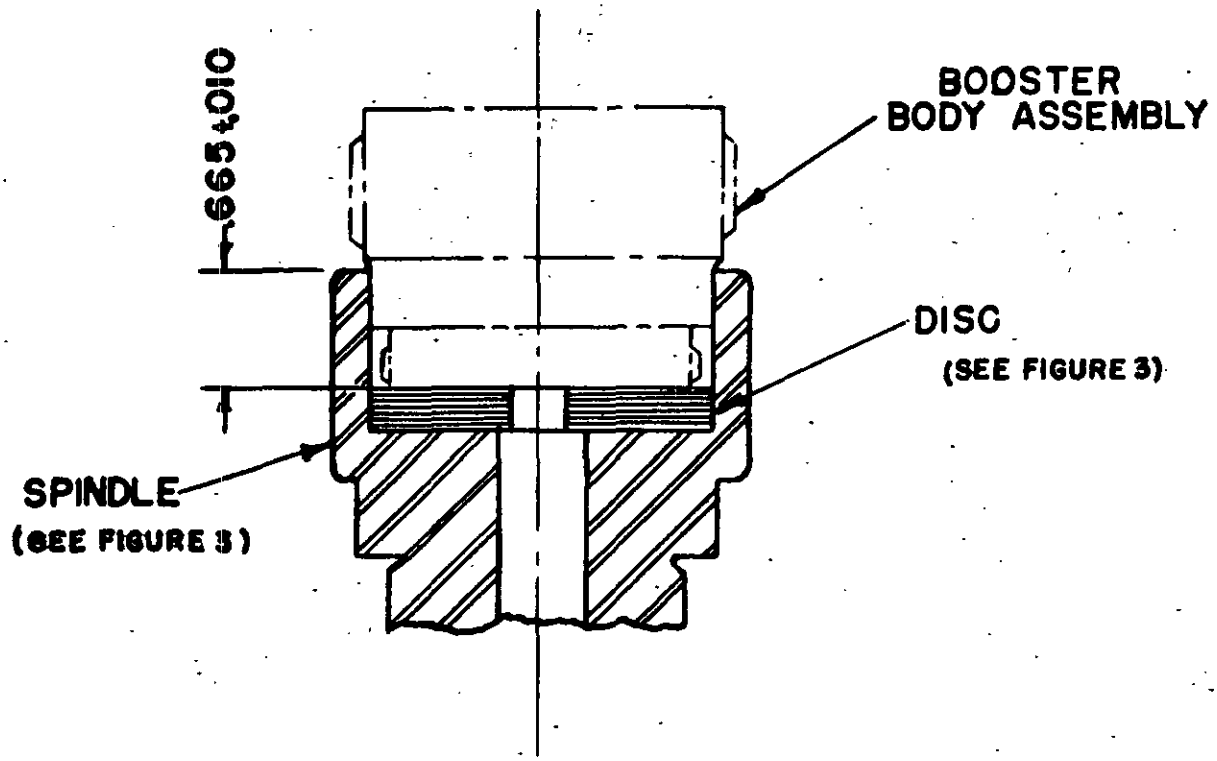
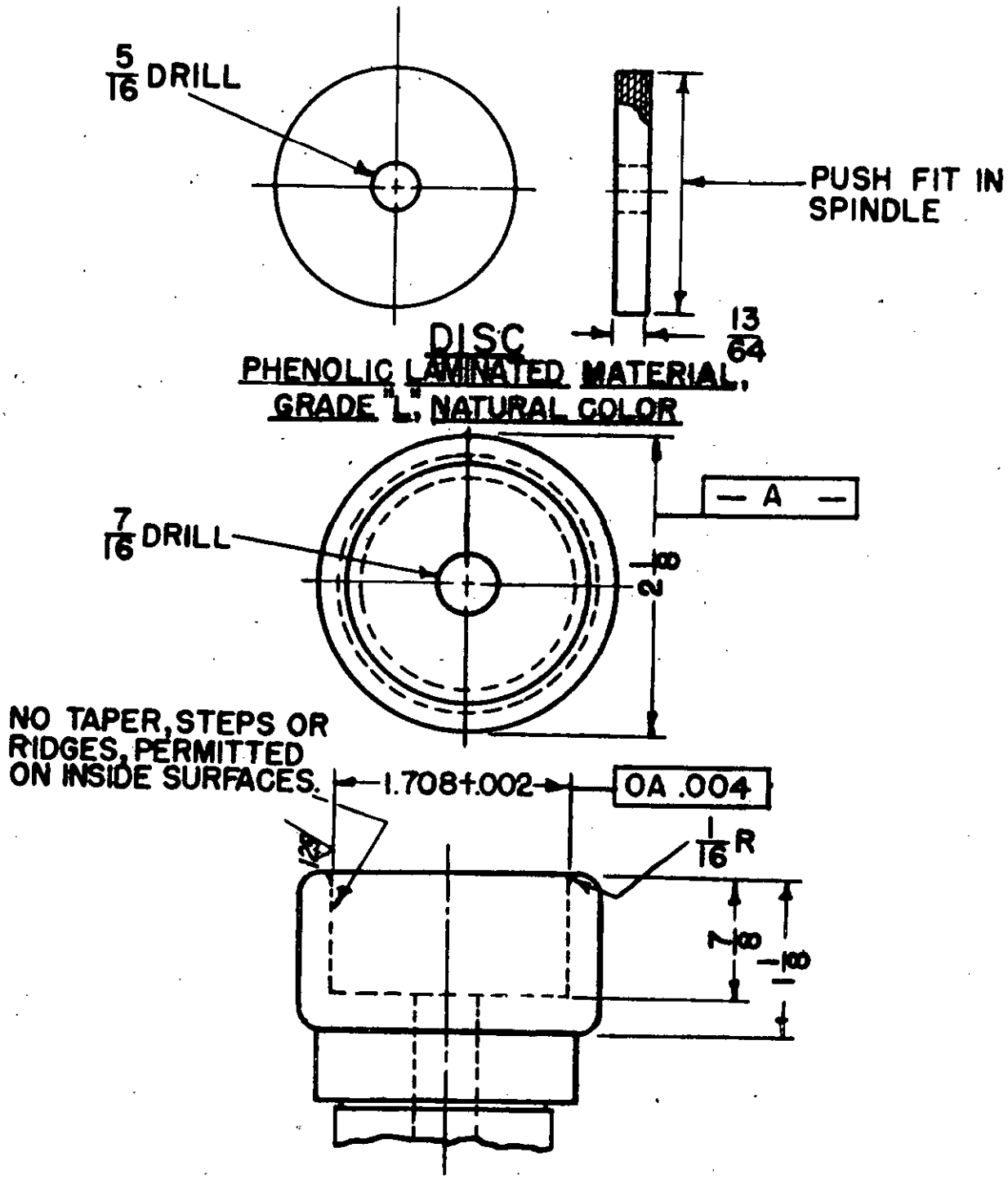


FIGURE 2

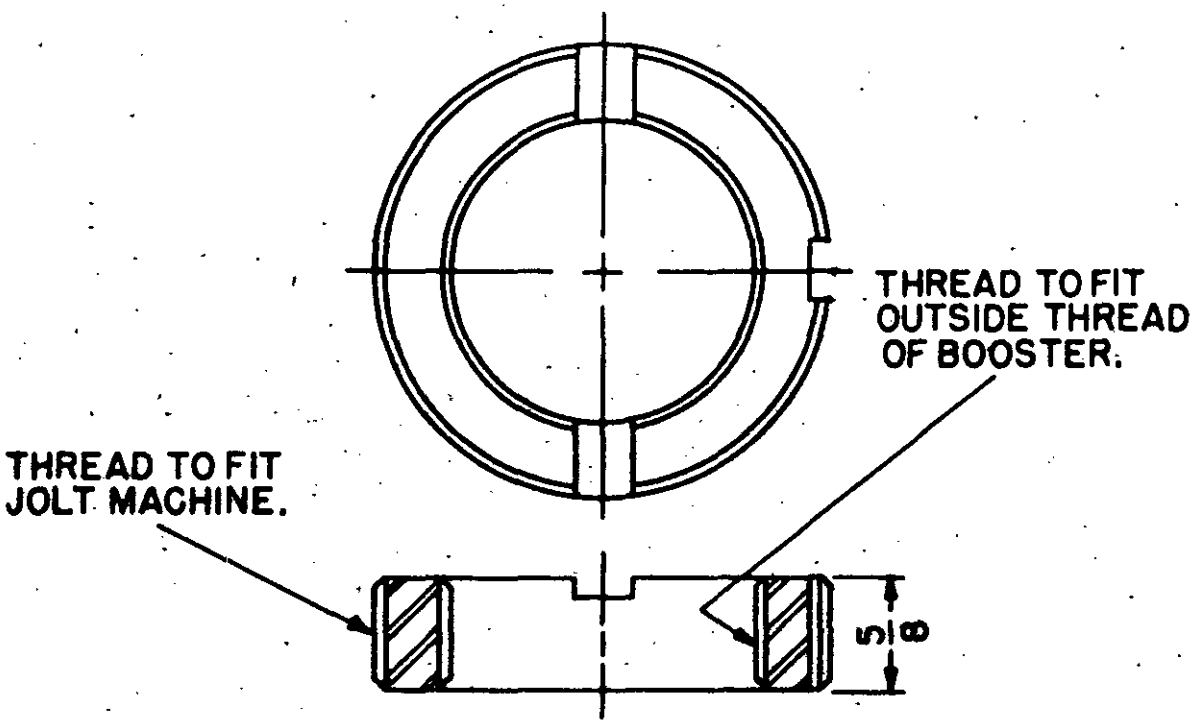
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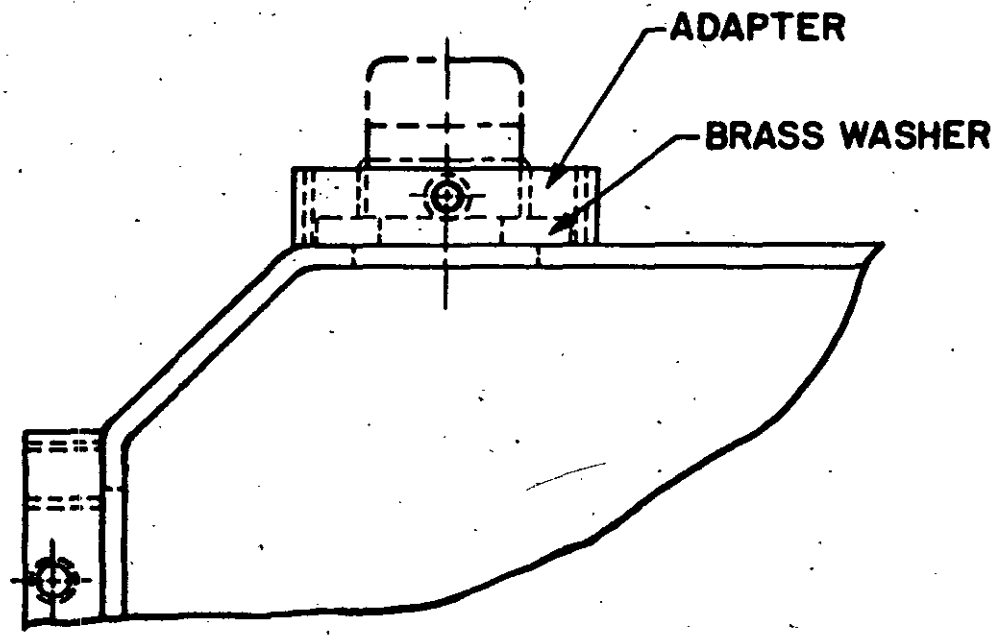
SPINDLE
STEEL, PACK HARDEN

FIGURE 3

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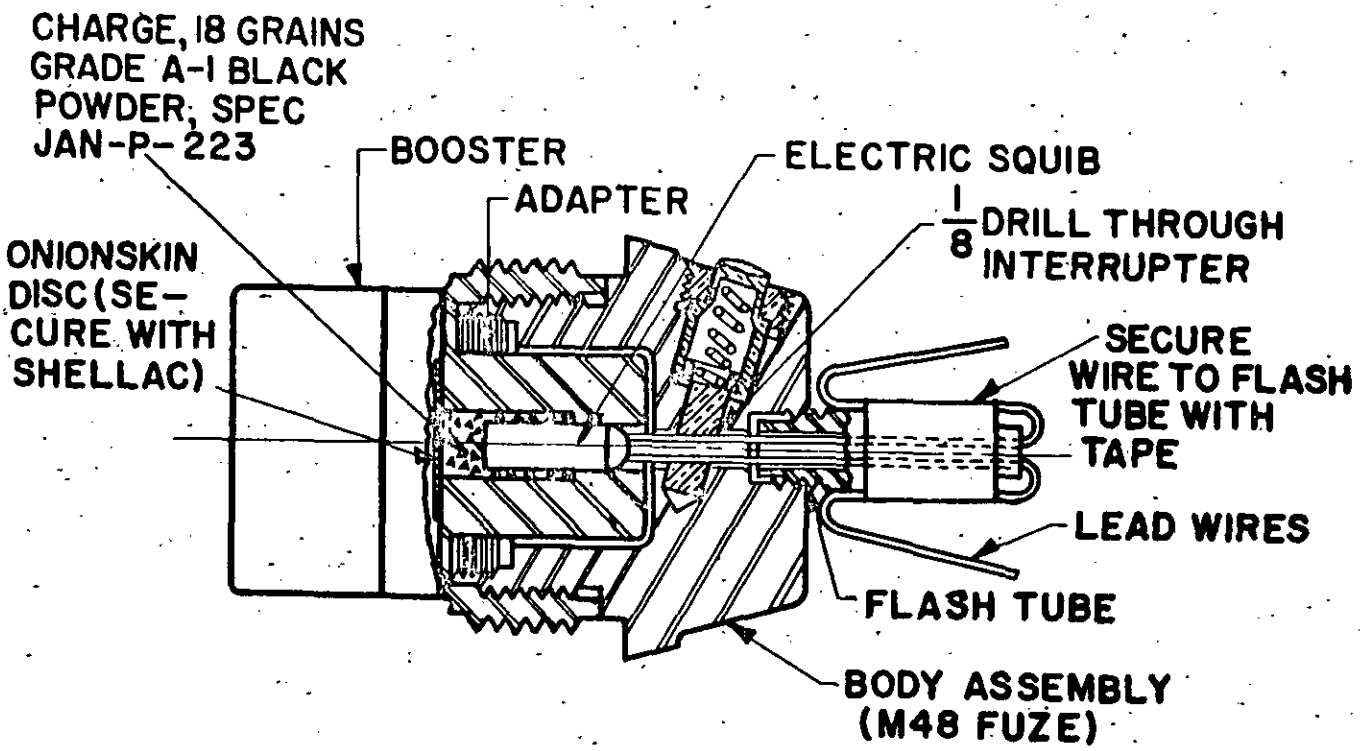
ADAPTER
STEEL



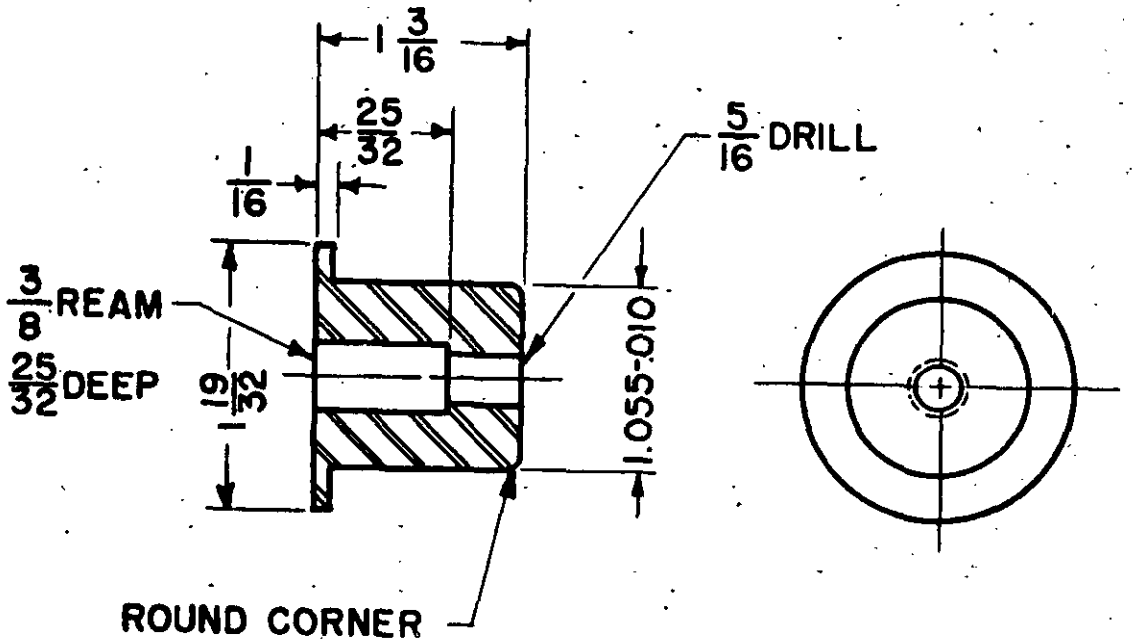
PART VIEW OF JOLT MACHINE.

FIGURE 4

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ASSEMBLY



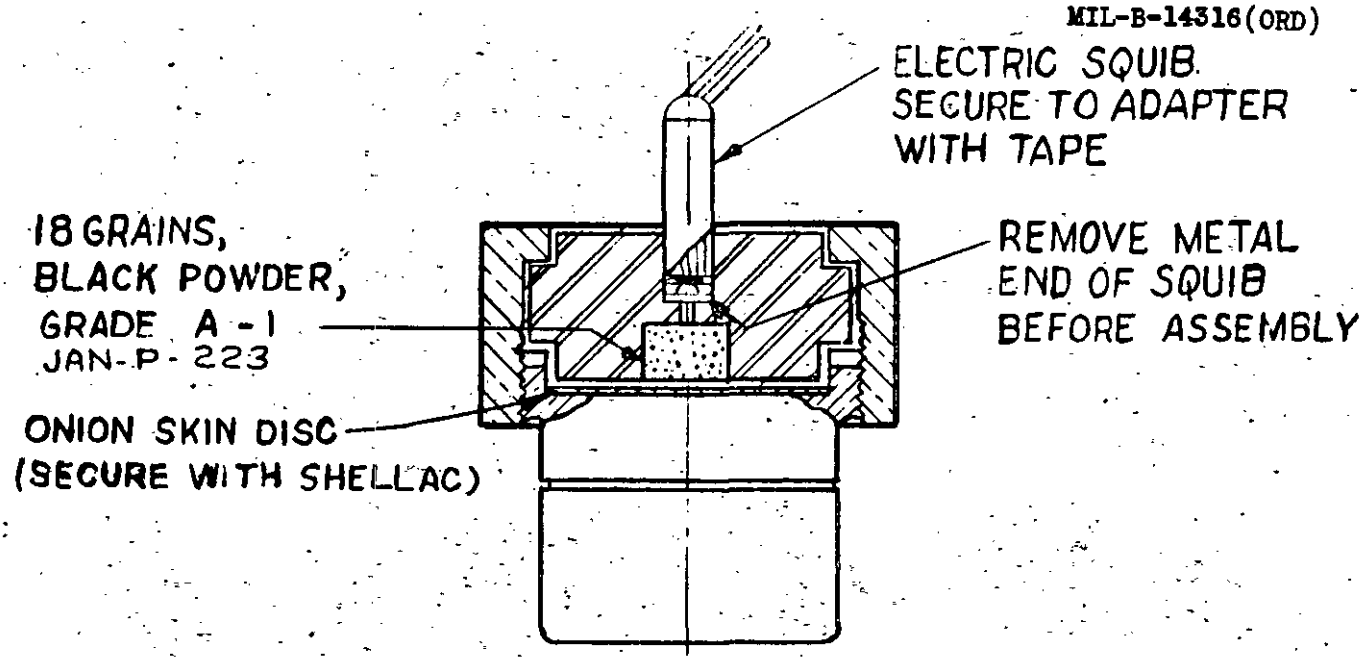
ADAPTER

COLD ROLLED STEEL

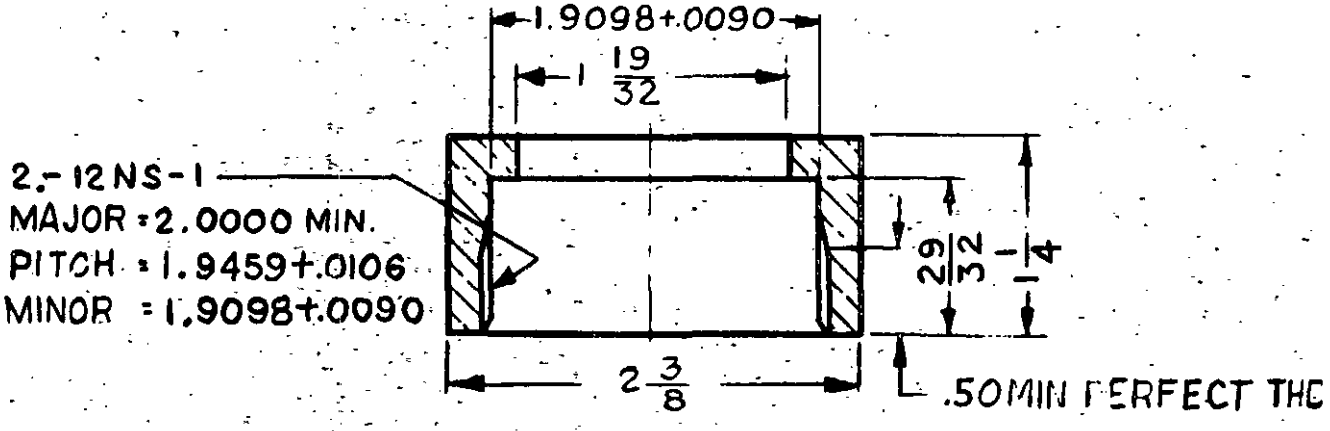
FIGURE 5

(APPLICABLE TO THE M21A4 BOOSTER)

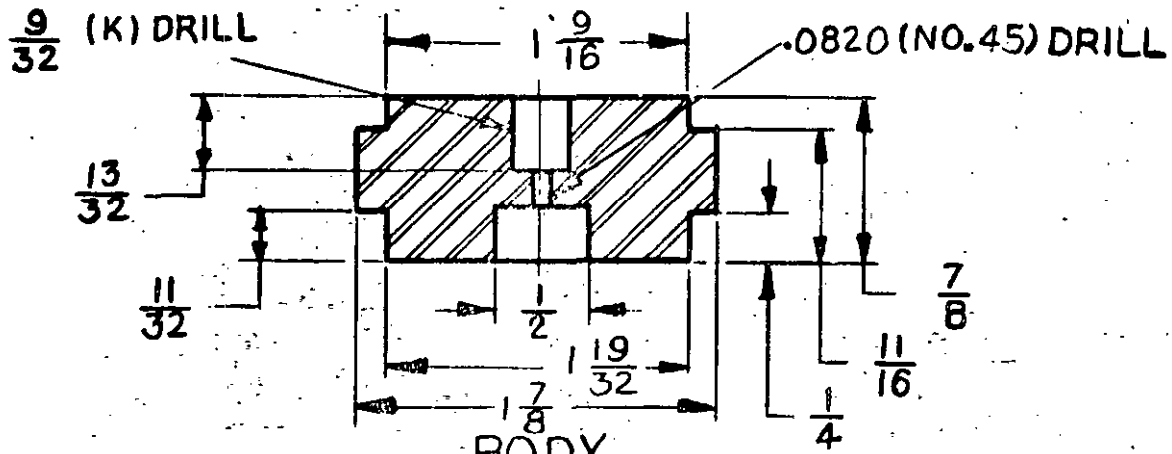
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ASSEMBLY



SLEEVE
BRASS OR STEEL, COMMERCIAL



BODY
BRASS OR STEEL, COMMERCIAL
(APPLICABLE TO THE M25 BOOSTER)

FIG. 6

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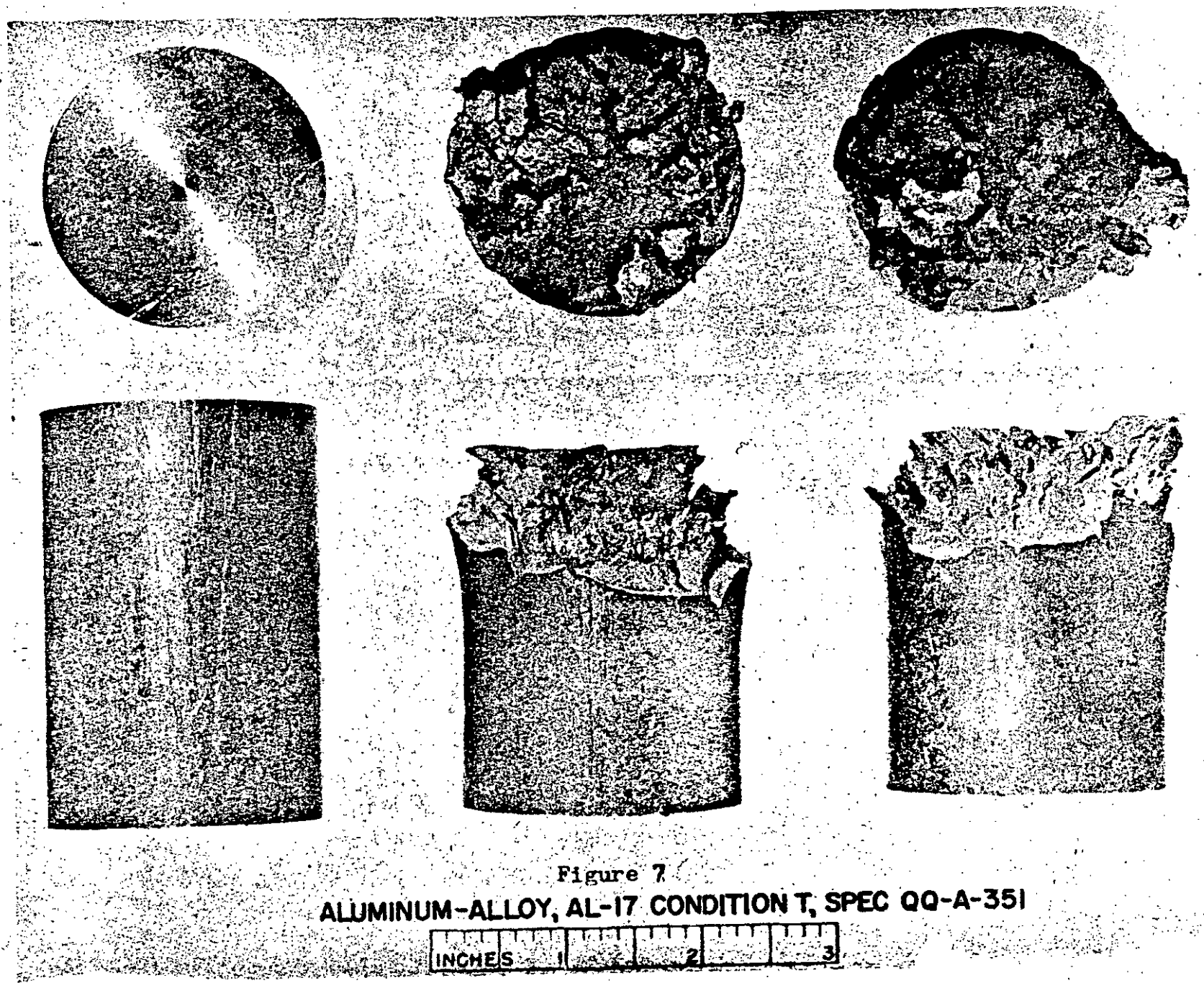


Figure 7

ALUMINUM-ALLOY, AL-17 CONDITION T, SPEC QQ-A-351

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